

Inflationary cosmology

Gytis Vėjelis

2025

Contents

1	Lecture 1. Is our universe part of a multiverse? I	2
2	Lecture 2. Is Our Universe Part of a Multiverse? II	6
3	The Dopler Effect and Special Relativity	8

1 Lecture 1. Is our universe part of a multiverse? I

The standard Big bang

What is it?

- Theory that the universe as we know it began 13-14 billion years ago. (Latest estimate: 13.82 ± 0.05 billion years)
- Initial state was a hot, dense, uniform soup of particles that filled space uniformly, and was expanding rapidly

What does it describe?

- How the early universe expanded and cooled
- How the light chemical elements formed (heavy elements formed in stars)
- How the matter congealed to form stars, galaxies, and clusters of galaxies

What is **doesn't** describe?

- What caused the expansion? (The big bang theory described only the aftermath of the bang.)
- Where did the matter come from? (The theory assumes that all matter existed from the beginning, again, it assumes that for every particle we see today there was at least a precursor particle if not the same particle today)

“In other words, it says nothing about what banged, why it banged or what happened before it banged! It's a bangless theory” - Alan Guth

Cosmic Inflation

Inflation is a modification of the standard big bang theory, providing a very brief “prequel”. Inflation can explain the **bang** of the big bang, it explains the outward propulsion in terms of GRAVITATIONAL REPULSION!

The combination of a general relativity and modern particle theories predict that, at very high energies, there exists forms of matter that create a gravitational repulsion! (In general relativity, gravitational repulsion is created by negative pressures).

Inflation proposes that a patch of repulsive gravity material existed in the early universe - for inflation at the grand unified theory scale ($10^{16} GeV$), the patch needs to be only as large as $10^{-28} cm$. (Since any patch is enlarged fantastically by inflation, the initial density or probability of such patches can be very low.) Note that $1 GeV$ is around the mass of a proton.

The gravitational repulsion created by this material was the driving force behind the big bang. The repulsion drove it into exponential expansion, doubling in size every $10^{-37} s$ or so! The patch expanded exponentially by a factor of at least 10^{28} (100 doublings), but it could have expanded much more. Inflation lasted maybe 10^{-35} second, and at the end, the region destined to become the presently observed universe was about the size of a marble.

The repulsive-gravity material is unstable, so it decayed like a radioactive substance, ending inflation. The decay released energy which produced ordinary particles, forming a hot, dense “primordial soup.” Standard cosmology began.

There is a caveat, though. The decay happens almost everywhere, but not everywhere. The inflaton decays like a radioactive substance, aka as an exponential with a halflife, however as long as you wait there is always some fraction of it remains. In some sense, inflation never really ends.

The density of the repulsive gravity material was not lowered as it expanded! It expands at a constant density. Although more and more mass/energy appeared as the repulsive-gravity material expanded, total energy was conserved!

The energy of a gravitational field is negative!

The positive energy of the repulsive gravity material was compensated by the negative energy of gravity. The TOTAL ENERGY of the universe may very well be zero.

Evidence for Inflation

1. **Large Scale uniformity.** The cosmic background radiation is uniform in temperature to one part in 100000. It was released when the universe was about 400000 years old. In standard cosmology without inflation, a mechanism to establish this uniformity would need to transmit energy and information at about 100 times the speed of light. Note that you went as measured CMB you would find a larger asymmetry than this. With one direction being hotter than the other direction. We interpret this asymmetry as our motion through the CMB (which makes it look hotter in one direction and colder in the other). This motion has a definite angular pattern. We determine our angular velocity from this asymmetry and subtract that out. This is the residual asymmetry.

Inflationary Solution: In inflationary models, the universe begins so small that uniformity is easily established - just like the air in the lecture hall spreading to fill it uniformly. Then inflation stretches the region to be large enough to include the visible universe.

2. **Flatness problem.** Why was the early universe so flat? Flat means Euclidean, as opposed to the non-Euclidean curved spaces that are also allowed by Einstein's general relativity. According to general relativity, the flatness of the universe is related to its mass density:

$$\Omega = \frac{\text{actual mass density}}{\text{critical mass density}},$$

where the "critical density" depends on the expansion rate. $\Omega = 1$ is flat, Ω greater than 1 is closed, Ω less than 1 is open. A universe at the critical density is like a pencil balancing on its tip. If Ω in the early universe was slightly below 1, it would rapidly fall to zero - and no galaxies would form. If Ω was slightly greater than 1, it would rapidly rise to infinity, the universe would recollapse, and no galaxies would form. To be as close to critical density as we measure today, at one second after the big bang, Ω must have equal to one to 15 decimal places.

Inflationary solution: Since inflation makes gravity become repulsive, the evolution of Ω changes, too. Ω is driven towards one, extremely rapidly. It could begin at almost any value. Since the mechanism by which inflation explain the flatness of the early universe almost always overshoots, it predicts that even today the universe should have a critical density. Latest observation by Planck satellite (combined with other astronomical observations)

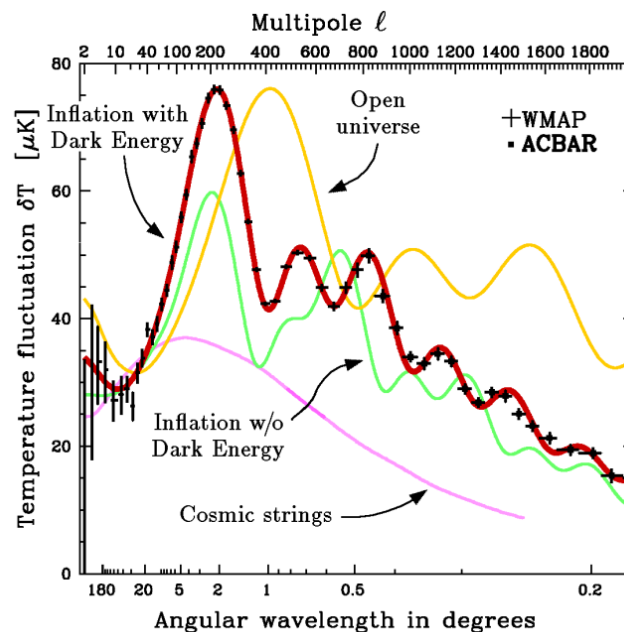
$$\Omega = 1.0010 \pm 0.0065$$

New ingredient: Dark Energy. In 1998 it was discovered that the expansion of the universe has been accelerating for about the last 5 billion years. The "Dark Energy" is the energy causing this to happen.

3. **Small scale nonuniformity.** Can be measured in the cosmic background radiation. The intensity is almost uniform across the sky, but there are small ripples. Although these ripples are only at the level of 1 part in 100000, these non-uniformities are now detectable! Where do they come from?

Inflationary solution: Inflation attributes these ripples to quantum fluctuations. Inflation makes generic predictions for the spectrum of these ripples (i.e. how the intensity varies with wavelength). The data measured so far agree beautifully with inflation. This like the Earth form because these ununiformities in the mass density are gravitationally unstable. In the standard Big Bang there is no justification for something like that, these mechanisms were put in "by hand".

And we see these non-uniformities in the CMB.



In inflation you cannot predict the amplitude of these fluctuations, since that would require a deeper understanding of high energy physics, however you can compute the spectrum.

Universe to Multiverse

The repulsive gravity material that drives the inflation is metastable. In any one location, the probability of remaining in an inflating state decreases with time - usually exponentially. But, the Universe in the meantime is expanding exponentially. In any successful version of inflation, the exponential expansion is faster than the exponential decay! Therefore, the volume that is inflating increases with time, even though the inflating material is decaying! The inflation becomes eternal -

once it starts, it never stops. The inflating region never disappears, but pieces of it undergo decay and produce "pocket universes," ad infinitum. Instead of one Universe, inflation produces an infinite number - **a multiverse**.

Dark Energy: Key Mystery of the Universe

In 1998, astronomers discovered that the Universe has been accelerating for about the last 5 billion years (out of its 14 billion year history).

Implication: Inflation is happening today, so the Universe today is filled with a repulsive gravity material. (Within general relativity, this requires negative pressure.) The repulsive gravity material, which apparently fills space, is called the "DARK ENERGY". Nobody knows what dark energy is, but the simplest explanation is that its vacuum energy, also known as a cosmological constant.

The quantum vacuum is far from empty, so a nonzero energy density is no problem. In quantum field theory, the energy density of quantum fluctuations diverges. All wavelengths contribute, and there is no shortest wavelength. A plausible cutoff for the fluctuations is the Planck length, around $10^{-33}cm$, the scale of quantum gravity. Using this cutoff, the estimated vacuum energy density is too large. However it is too large by 120 orders of magnitude! lol

The Landscape of String Theory

Since the inception of string theory, theorists have sought to find the vacuum of string theory - with no success. Within the past 20 years or so, most string theorists have come to the belief that there is no unique vacuum. Instead, there are maybe 10^{500} long-lived metastable states, any of which could serve as a substrate for a pocket Universe. This is the Landscape!

Eternal inflation can presumably produce an infinite number of pocket universes of every type, populating the landscape. Although string theory would govern everywhere, each type of vacuum would have its own low-energy physics - its own "standard model", its own "constants" of nature, and its own vacuum energy density.

2 Lecture 2. Is Our Universe Part of a Multiverse? II

Summary of Last Lecture

The Standard Big Bang: Really describes only the aftermath of a bang, beginning with a hot dense uniform soup of particles filling an expanding space.

Cosmic Inflation: The prequel, describes how repulsive gravity - a consequence of negative pressure - could have driven a tiny patch of the early Universe into exponential expansion. The total energy would be very small or maybe zero, with the negative energy of the cosmic gravitational field canceling the energy of matter:

1. Inflation can explain the large-scale uniformity of the Universe. (Cosmic microwave background (CMB) uniform to 1 part in 100000)
2. Inflation can explain why $\Omega \equiv \rho/\rho_{\text{crit}} = 1$ was accurate to > 15 decimal places at $t = 1$ seconds. Predicts $\Omega = 1$. Data $\Omega = 1.0010 \pm 0.0065$.
3. Predicts small quantum fluctuations in the mass density, which can be seen today as ripples in the CMB. Predictions agree very well with data.

Most inflationary models become eternal - the expansion overpowers the decay of the repulsive gravity material, so inflation never ends. An exponential growing and never-ending number of pocket universes are formed where decays occur. The expansion of the Universe is accelerating, indicating that space is filled with "dark energy", most simply described as vacuum energy. Vacuum energy in a quantum field theory is not surprising - field fluctuations, nonzero Higgs field 0 there are positive and negative contributions. But typical magnitudes are 10^{120} times too large.

The landscape of String Theory: String theory predicts 10^{500} long-lived, metastable "vacua", any one of which can act as the vacuum for a pocket Universe. Each would have its own value for the vacuum energy density, with values ranging from roughly -10^{120} to $+10^{120}$ times the observed value. If the landscape has 10^{500} vacua, and a fraction 10^{-120} have small vacuum energy densities like our universe, then we expect about

$$10^{-120} \times 10^{500} = 10^{380}$$

vacua with low energy densities like ours. How could we explain why we are living in such a fantastically unusual type of vacuum? Consider, as an example, the local density of matter in which we find ourselves - is about 10^{30} times larger than the mean density of the visible Universe. Most of us would presumably accept this as a selection effect: life can evolve only in those rare regions of the Universe where the density of matter is unusually high. As early as 1987, Steve Weinberg pointed out that the vacuum energy density might be explained in the same way. Maybe the vacuum energy density is huge in most pocket universes. Nonetheless, we need to remember that vacuum energy causes the expansion of the Universe quickly collapses. If large and positive the universe flies apart before galaxies can form. It is plausible, therefore, that life can arise only if the vacuum energy density is very near zero.

In 1998 Martel, Shapiro and Weinberg made a serious calculation of the effect of the vacuum energy density on galaxy formation. They found that to within a factor of order 5, they could "explain" why the vacuum energy density is as small as what we measure.

Some consider these anthropic arguments as ridiculous. The anthropic explanation (for anything) should be considered the explanation of last resort. Until we actually understand the landscape, and the initiation of life, we can only give plausibility arguments for anthropic explanations. Hence the

anthropic arguments only become attractive when the search for more deterministic explanations has failed, as so far is the case for the vacuum energy density.

3 The Dopler Effect and Special Relativity

Consider the case of a stationary observer and a source that is moving away with some velocity v i.e. the velocity of recession, ΔS is the period of wave at source, Δt_O is the period of the wave at observer. The source emits a wave in some medium which travels at speed u . We want the relationship between the t 's.

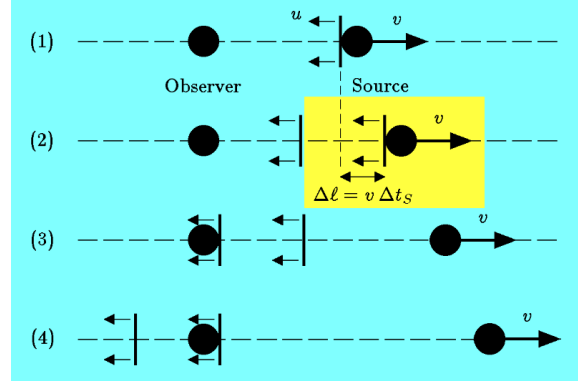


Figure 1: Non-relativistic case. Source moves away from observer.

- (1) The source is emitting a wave crest
- (2) the source emits a second wave crest, but it also has moved. The highlighted part shows that the second wave crest has to move a bit more than the first one. This is what accounts for the Doppler effect. Here Δl is the crucial quantity
- (3) The waves have traveled and the first crest has hit the observer
- (4) The second crest has hit the observer.

To figure out the Doppler shift, realise that if both objects were stationary then their periods would coincide i.e. $t_O = t_S$. But since there is motion, the second crest will have to travel further by Δl , so

$$\Delta t_O = \Delta t_S + \frac{\Delta l}{u} = \Delta t_S + \frac{v \Delta t_S}{u} \Rightarrow \frac{\Delta t_O}{\Delta t_S} = 1 + \frac{v}{u} = \frac{\lambda_O}{\lambda_S} = 1 + z$$

where λ_S and λ_O are the wave wavelengths for source and the observer correspondingly, z is **the redshift**. Note that $z = 0$ corresponds to stationary situation i.e. no redshift. In this case

$$z = \frac{v}{u} \quad \text{Nonrelativistic, source moving}$$

Now let's consider a different case, where the observer is moving with velocity v and the source is stationary.

- (1') Again, first wave crest is emitted
- (2') The second wave crest is emitted by the source

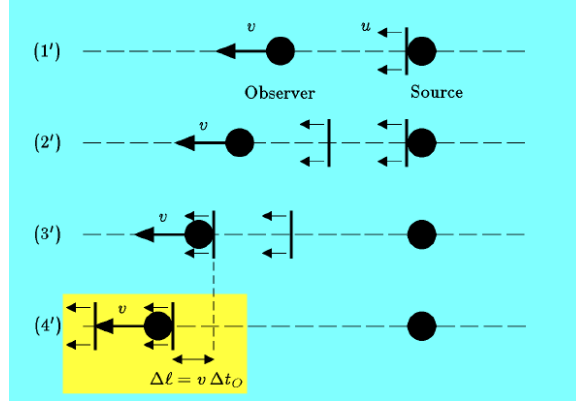


Figure 2: Non-relativistic case. Observer moves away from the source

(3') The first wave crest arrives at the source

(4') The second wave crest arrives at the source. In this case this is the most important frame. At the moment the second wave crest reaches the observer, the observer has moved by a distance Δl

So similarly, if there was no motion then $\Delta t_O = \Delta t_S + \frac{\Delta l}{u} = \Delta t_S + \frac{v \Delta t_O}{u}$. Solving this equation we obtain

$$\frac{\Delta t_O}{t_S} = \left(1 - \frac{v}{u}\right)^{-1}$$

Hence the redshift

$$z = \frac{\Delta t_O}{\Delta t_S} - 1 = \left(1 - \frac{v}{u}\right)^{-1} - 1 = \frac{v/u}{1 - v/u} \quad \text{Nonrelativistic, observer moving}$$

Note that

$$z_{\text{O moving}} - z_{\text{S moving}} = \frac{(v/u)^2}{1 - (v/u)}$$

Note that this shows more explicitly that z is proportional to the square of v/u . If we had a small v/u then the denominator would vanish. Also note that this does not break the galilean relativity, but when we transform into another frame we get that the air is moving i.e. when we say that something is at rest, it is at rest to the medium that is transporting the wave. Let's move to the relativistic case.

We will consider three effects of special relativity: **Time dilation**. Any clock which is moving at speed v relative to a given frame will "appear" (to an observer using that reference frame) to run slower than normal by a factor γ , and given by

$$\gamma \equiv \frac{1}{\sqrt{1 - \beta}}, \quad \beta \equiv v/c$$

Notice that for $v \ll c$ the effect is small and $\gamma \simeq 1$, meaning that effectively there is no effect. The word "appear" is a subtle term. When we say it, we don't mean that we literally see the clock

running slower, since seeing just means that we're measuring the light pulses as they arrive with our eyes and since light has finite travel time, that means that we measure things at different times. Suppose a large object is coming towards you, you would be seeing the back at an earlier time (the light travels longer) than the front. So the problem is that we're seeing different pieces of it at different times, in terms of the actual existence of the thing. So in order to know what we're **seeing** we would need to calculate point by point what you would be seeing for every part of the object. The simple expressions like the one above are based on what a **frame of reference sees**. And a frame of reference we consider as a structure of ruler and with clocks located everywhere in that grid (all of which have been synchronized to start with). And if you want to know at what time an event happens you look at the clock next to it. And when we compare how the local clocks would compare, we see this time dilation effect (taking out the light travel time). In regards to the relativistic doppler shift, we'll see that the clock coming towards the observer is running **faster** and this happens because the distance gets smaller beating the effect of time dilation. To recover regular time dilation we would need moving clocks speed be perpendicular to the observer, meaning that the photon going from the clock would need to be perpendicular to the travel velocity of the clock wrt the observer.

Now, doing the calculation we assume that the wave is a light wave and the velocities are comparable to c i.e. time dilation effect is important enough and we want to take it into account. Now we would expect to get the same results, regardless if the observer or the source is moving. Since previously, it was the air that was not stationary anymore when we transformed into another reference frame.

It is crucial that what we talk about things measures from a particular reference frame, since the transformation from one reference frame to another is non-trivial. So the time between the pulses as measured from the source clock is what we call Δt_S . The source (effectively a clock since it's emitting pulses at regular intervals which fits in the definition of a clock) is moving in this picture, so relative to our frame (again we always think of properties as being consistent in our frame), it's clock will be running slower by a factor of gamma. And since the observer is not moving there is no time dilation associated with the observer.

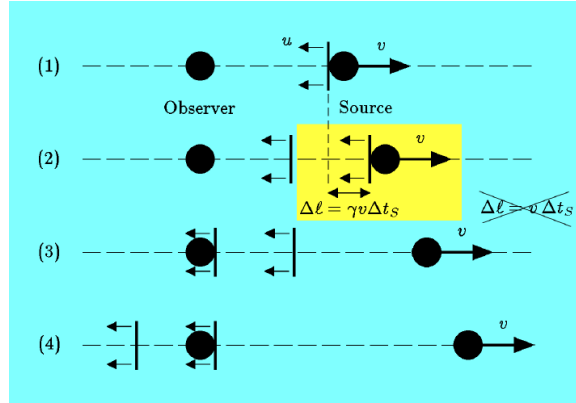


Figure 3: Relativistic case. Source moves away from the observer

$$\Delta t_O = \gamma \Delta t_S + \frac{\Delta l}{u} = \gamma \Delta t_S + \frac{v \gamma \Delta t_S}{u} = \gamma \left(1 + \frac{v}{c} \right) \Delta t_S$$

but note the difference of the squared in γ factor.

$$\Delta t_O = \sqrt{\frac{1+\beta}{1-\beta}} \Delta t_S \quad \text{Relativistic, source moving}$$

Next, we'll do the relativistic case of observer moving. Our frame of reference is the frame of reference of the image, so we'll see the observers clock running slower.

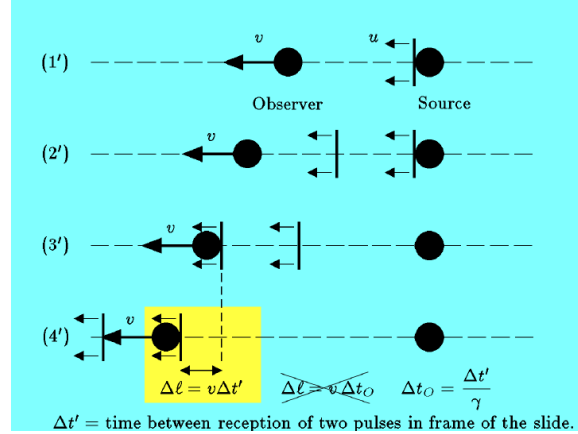


Figure 4: Relativistic case, observer is moving

Again, we want everything in terms that we're measuring it. So the distance the observer moved is its velocity times the time elapsed from frame (3') to frame (4'). This time our reference frame coincides with the reference frame of the source.

$$\begin{aligned} \Delta t' = \gamma t_O &\Rightarrow \Delta t' = \Delta t_S + \frac{v \Delta t'}{c} \Rightarrow \Delta t' = \left(1 - \frac{v}{c}\right)' \Delta t_S \\ \Rightarrow \Delta t_O = \frac{1}{\gamma} \left(1 - \frac{v}{c}\right)^{-1} \Delta t_S &= \sqrt{(1+\beta)(1-\beta)} \frac{1}{1-\beta} \Delta t_S \end{aligned}$$

$$\Delta t_O = \sqrt{\frac{1+\beta}{1-\beta}} \Delta t_S \quad \text{Relativistic, source or observer moving}$$

$$z = \frac{\Delta t_O}{\Delta t_S} - 1 = \sqrt{\frac{1+\beta}{1-\beta}} - 1 \quad \text{Relativistic}$$

Special relativity only describes things happening only in inertial reference frames (uniform velocity). What about clocks that are accelerating? As long as gravity is absent we can use the dynamical equations of special relativity. The clock is a complicated system so to simplify matters we assume that the acceleration does not affect its speed. The assumption that we're making is that these are ideal clocks (built well and does not break under large accelerations) and for the "acceleration does not affect its speed" part we assume that the clock runs the exactly same as the clock running along side it at the same speed but with no acceleration. At every particular point during the motion of the clock the tick speed is effected by the motion by a factor of gamma, but

we're assuming that it is the same as the effect of non-accelerating clock, meaning that there is no effect of acceleration.

Kinematic effects (3 and no more), are time dilation, which we have already seen. Second, is the Lorentz (-Fitzgeralds) contraction: any rod which is moving at a speed v along its length relative to a given reference frame will "appear" (to an observer using that reference frame) to be a shorter than its normal length by the same factor γ . A rod which is moving perpendicular to its length does not undergo a change in apparent length.

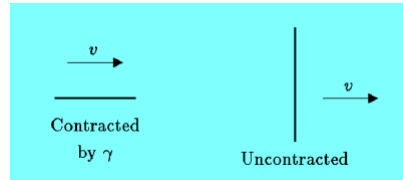


Figure 5: Lorentz contraction

Next is the relativity of simultaneity: suppose a rod which has rest length l_0 is equipped with a clock at each end. The clocks can be synchronized in the rest frame of the system by using light pulses. (That is, a light pulse can be sent out from the center, and the clocks at both ends can be started when they receive the pulses.) If the system moves at speed v along its length, the trailing clock will "appear" to read a time which is later than the leading clock by an amount $\beta l_0/c$. If, on the other hand, the system moves perpendicular to its length, then the synchronization of the clocks is not disturbed.

This is a crucial concept. Since on the first look it might seem like two observers moving relative to each other measure other's clock as ticking slower, however the concept of simultaneity would show that the clocks one reference frame would not be considered synchronised.